NodeJS

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Agenda

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02 Steps in Writing Event Driven Code

03 EventEmitter

04 Registering Callbacks

05 Emitting Events

06 Emit and Respond to Events

07 Benefits of Event Driven Programming

08 Streams

09 Types of Streams

10 Read Data from a file using streams

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Agenda

11 Write Data to a file using streams

12 Stream Events

13 Process Stream Events

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Express JS

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What is ExpressJS?

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What is ExpressJS?

ExpressJS is Web Application Development Framework based on Node.js

It takes care of a lot of common tasks that you would need to perform if you were to build a web application in Node JS, such as parsing the request, routing, error handling etc.

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Why ExpressJS?

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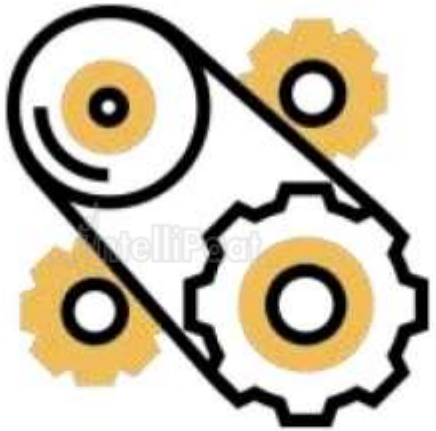
Why ExpressJS?

Express JS provides several benefits

- Fast
- Minimal
- Un-Opinionated
- Simple
- Unobtrusive

ExpressJS is a small framework and does not slow down an application



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Why ExpressJS?

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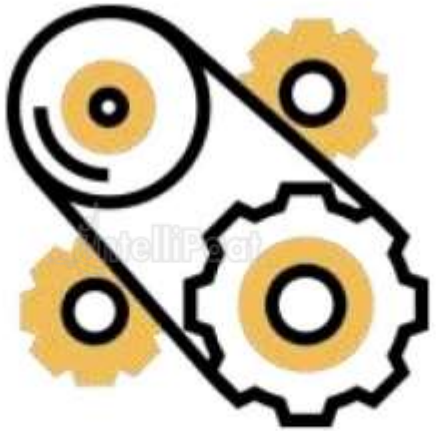
ExpressJS is a small framework and only provides fundamental web application features



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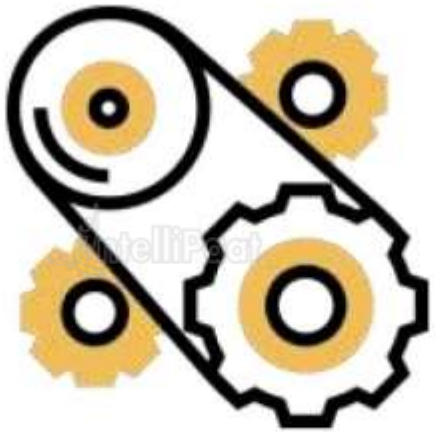
ExpressJS does not have default configurations, so you can structure your application however you like



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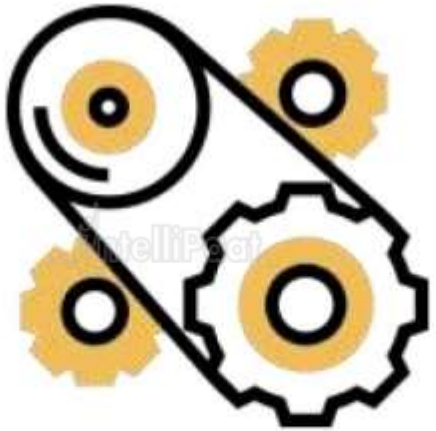
ExpressJS is very simple and easy to use



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Why ExpressJS?

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- Fast
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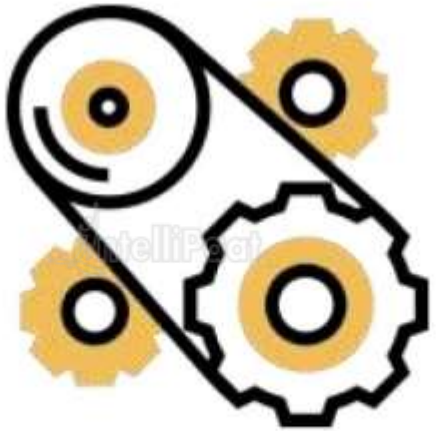
You can setup ExpressJS however you like and use it with any library of your choice



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Installing ExpressJS

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Installing ExpressJS

Since we are using NodeJS we will need to install ExpressJS using NPM

To install ExpressJS in a node application firstly generate an NPM project using the command npm init –y

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Installing ExpressJS

After generating the project run the command npm install express --save

Make sure that you are using the –save flag and not the –save-dev flag as we would need ExpressJS for our application to run

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ExpressJS Components

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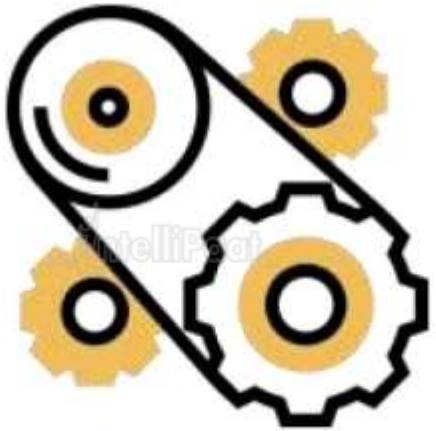
ExpressJS Components

ExpressJS has Several Components

- Request
- Response
- Middleware
- Error Handling
- Templates
- Database

ExpressJS builds a request object automatically, which gets populated with important data, like method, headers and so on



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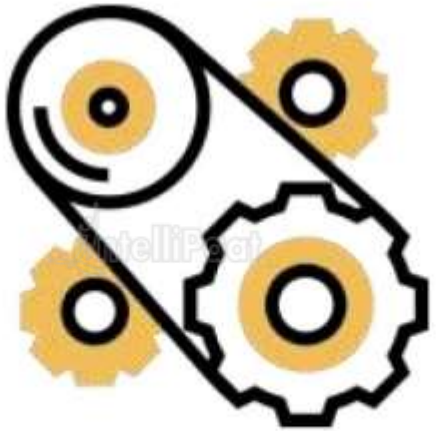
Express also builds a Response object which is an HTTP response that gets sent back to the user



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ExpressJS Components

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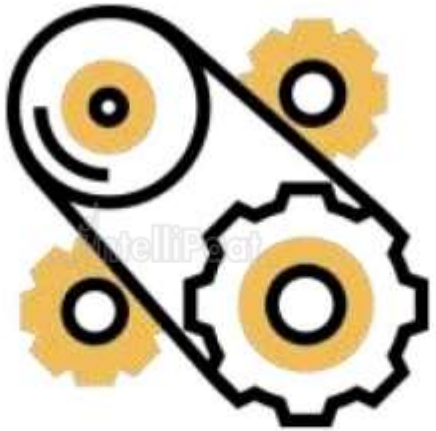
A Middleware is just a function that gets called during the Request – Response cycle and has access to request and response objects



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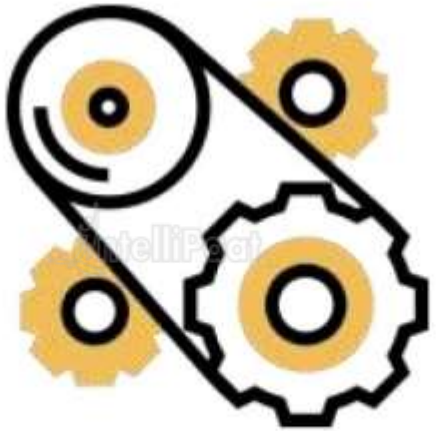
Express comes with a default error handler which will catch any synchronous or asynchronous error



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ExpressJS Components

ExpressJS has Several Components

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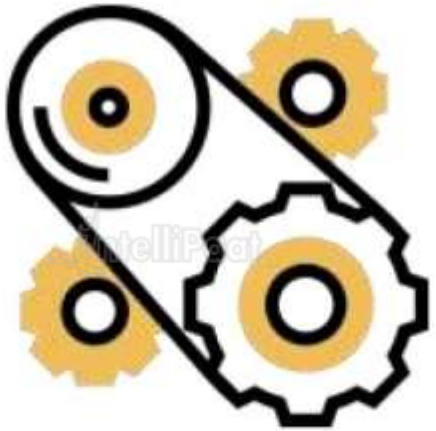
Since ExpressJS is unopinionated we can use any templating language that we wish to, e.g. ejs, jade etc



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ExpressJS Components

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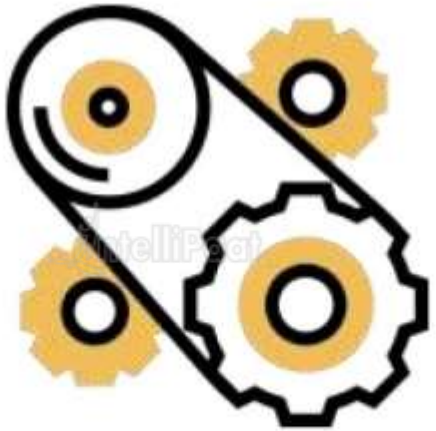
You can connect to any database using your ExpressJS app by loading an appropriate Node.js driver for the database



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HTTP Routing in ExpressJS

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HTTP Routing in ExpressJS

Express comes with an in-built router that allows developers to write code that handles request to a particular route effectively

In express the router has methods to respond to any or all of the seven HTTP methods (GET, POST, PUT, PATCH, DELETE etc). Each method is used to perform a specific action on the server

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HTTP Request Methods

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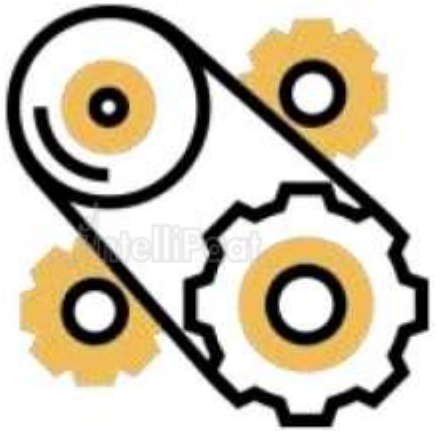
HTTP Response Classes

There are seven HTTP request methods

- GET
- POST
- PUT
- PATCH
- DELETE
- OPTIONS
- HEAD

GET Method is used to request server to return some data, either a collection or entity



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HTTP Response Classes

There are seven HTTP request methods

- GET
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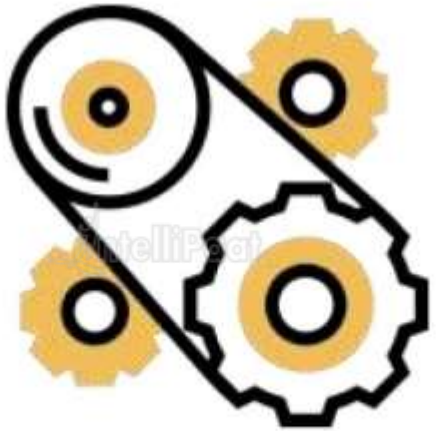
POST Method is used to submit data to the server, either for saving or further processing



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HTTP Response Classes

There are seven HTTP request methods

- GET
- POST
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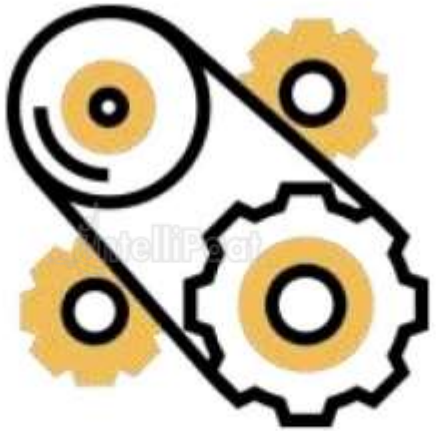
PUT method is used to send some data to completely replace an entity on the server



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HTTP Response Classes

There are seven HTTP request methods

- GET
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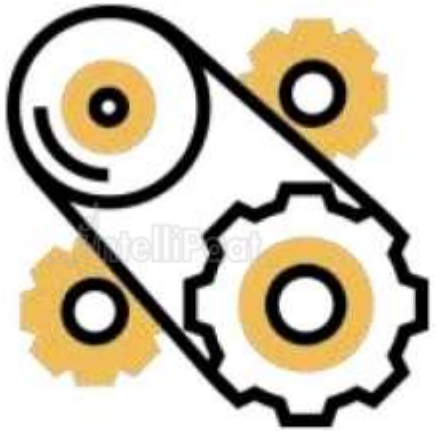
- PUT
- PATCH
- DELETE
- OPTIONS
- HEAD

PATCH method is used to send some data to partially replace an entity on the server



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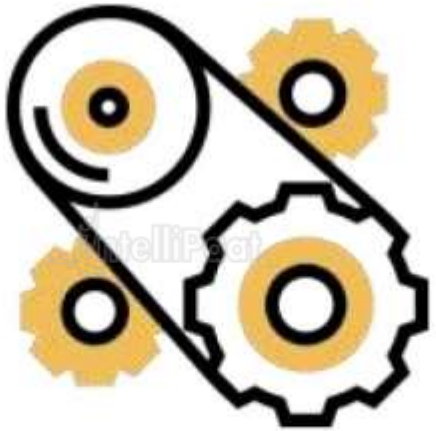
HTTP Response Classes

There are seven HTTP request methods

- GET
- POST
- PUT
- PATCH
- DELETE
- OPTIONS
- HEAD

DELETE method is used to delete an entity on the server



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HTTP Response Classes

There are seven HTTP request methods

- GET
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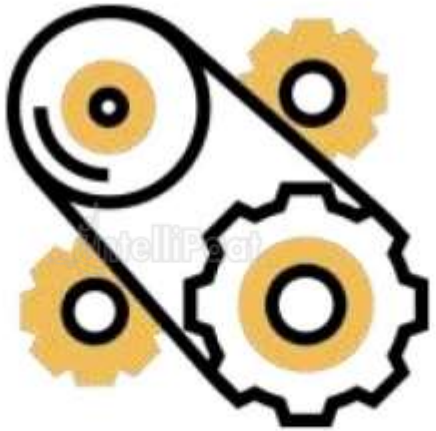
- OPTIONS
- HEAD

OPTIONS method is used to describe the communication options for the target resource



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HTTP Response Classes

There are seven HTTP request methods

- GET
- POST
- PUT
- PATCH
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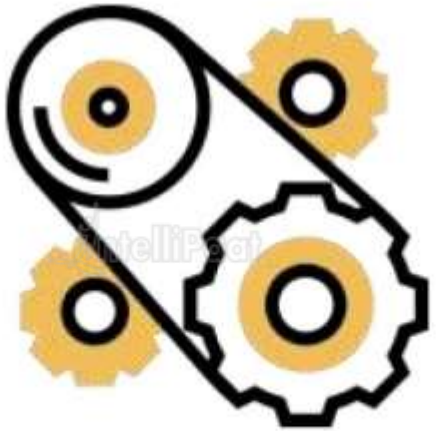
- OPTIONS
- HEAD

The HEAD method is sued to get a response that is same as that of a GET request, but without the response body



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Templating

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Templating

ExpressJS can be easily integrated with a lot of templating engines rather easily

Since express does not ship with a default templating engine you can either create your own or download a package that does the templating for you and use it with HTTP

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Templating Engines

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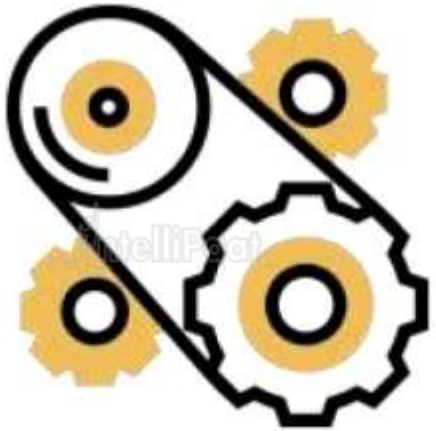
Templating Engines

There are many templating engines out there on NPM and can be integrated with ExpressJS. Here are some of the most popular ones

- EJS
- MustacheJS
- HandlebarsJS
- Nunjucks
- PUG

EJS stands for Embedded JavaScript. It is one of easiest templating engines



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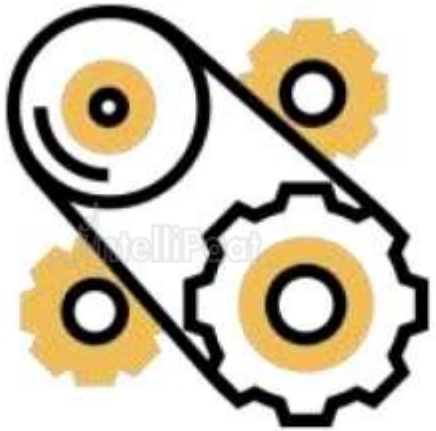
Mustache is one of the most popular templating systems that works for a number of programming languages, including JS, Node.js, etc.



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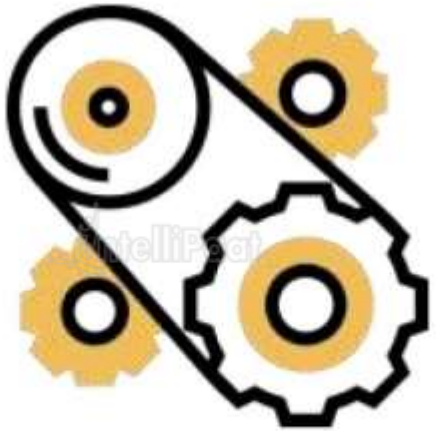
Handlebars is another popular templating language that is a superset of MustacheJS with some additional features



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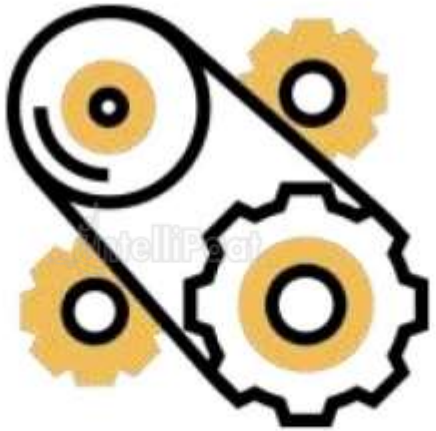
Nunjucks is developed by Mozilla and is an extremely powerful templating language. It's syntax is inspired by python's Jinja2 Templates



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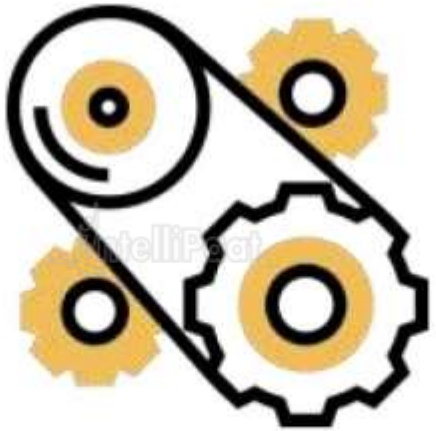
- PUG

PUG is the most popular templating language among the Express community. It was previously known as JADE



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Quiz

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Quiz

1. Which method is used to read file synchronously?

A	readFileSynchronously
B	readFileSync
C	readFS
D	readFile

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Quiz

1. Which method is used to read file synchronously?

A	readFileSynchronously
B	readFileSync
C	readFS
D	readFile

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Quiz

2. Which method is used to read file asynchronously?

A	readFileAsynchronously
B	readFileAsync
C	readFAS
D	readFile

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Quiz

2. Which method is used to read file asynchronously?

A	readFileAsynchronously
B	readFileAsync
C	readFAS
D	readFile

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Quiz

3. Which constant is NodeJS is used to refer to absolute path of the current file?

A	__filename
B	__dirname
C	__absoule_file_path
D	__abs_filename

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Quiz

3. Which constant is NodeJS is used to refer to absolute path of the current file?

A	__filename
B	__dirname
C	__absoule_file_path
D	__abs_filename

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Quiz

4. Which constant is NodeJS is used to refer to absolute path of the current directory?

A	__filename
B	__dirname
C	__absoule_dir_path
D	__abs_directory

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Quiz

4. Which constant is NodeJS is used to refer to absolute path of the current directory?

A	__filename
B	__dirname
C	__absoule_dir_path
D	__abs_directory

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Quiz

5. Which property of the filestats object can be used to determine the time the file was last accessed?

A	access_timestamp
B	birthname
C	atime
D	uid

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Quiz

5. Which property of the filestats object can be used to determine the time the file was last accessed?

A	access_timestamp
B	birthname
C	atime
D	uid

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